Steve Nwaiwu

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Website Google Scholar ORCID



Employment History

2024 – 2025 **Graduate Teaching Assistant,** Rajamangala University of Technology Thanyaburi, Thailand.
Supported programming classes, led Python exercises, and assisted with assessment and student projects.

Jun 2023 – Sep 2023 **AI & Data Analyst,** Academic Resources Institute, RMUTT, Thailand.
Developed Python dashboards, analyzed institutional data, and automated reporting workflows.

2020 – 2023 **Web Developer,** Optice Tech LLC, Wyoming, USA.
Built and maintained web systems, integrated backend services, and improved performance and usability.

2016 – 2020 **Communications and Technology Department,** MTN Nigeria.
Supported communication systems, technical troubleshooting, and daily technology operations.

2013 – 2016 **Computer Science Teacher,** King's College, Lagos, Nigeria.
Taught Computer Science and robotics, helping students develop programming and problem-solving skills.

2012 – 2015 **Data and Information,** Guaranty Trust Bank, Lagos, Nigeria.
Analyzed transaction data, prepared operational reports, and supported reconciliation processes. Assisted with data quality checks and secure handling of customer information.

2012 – 2013 **Associate Data Protection and Privacy Officer (P-2),** United Nations (UNHCR), Nigeria.
Supported privacy risk assessments and secure handling of refugee data. Reviewed data-sharing practices and assisted with incident documentation and privacy guidance.

2011 – 2012 **IT Support Specialist,** Heritage Solutions & Technology, Lagos, Nigeria.
Provided IT support, maintained systems, assisted with infrastructure upgrades, and trained users.

Education

2026 – 2029 **Ph.D. in Artificial Intelligence,** Walailak University, Thailand.
Research focus: causal reasoning and interpretability in large language models for financial decision-making under distribution shift.

2024 – 2025 **M.Sc. (Research) in Data & Information Science,** Rajamangala University of Technology Thanyaburi, Thailand.
Thesis research: robust and explainable machine learning for hate-speech detection, with counterfactual reasoning, adversarial perturbations, and fairness-aware evaluation.

2020 – 2024 **B.Sc. (Honors) in Big Data Analytics & Management,** Rajamangala University of Technology Thanyaburi, Thailand.
Capstone: institutional analytics platform combining ETL pipelines, performance monitoring, and interactive visualization.

Education (continued)

2008 – 2011 **Diploma in Computer Science,** Yaba College of Technology, Lagos, Nigeria.

Research Publications

Journal Articles

1. S. Nwaiwu et al., "Decoding disinformation: A feature-driven explainable AI approach to multi-domain fake news detection," *Applied Sciences*, vol. 15, no. 17, 2025, Art. no. 9498. DOI: 10.3390/app15179498
2. S. Nwaiwu, "Parameter-efficient fine-tuning for low-resource text classification: A comparative study of LoRA, IA3, and ReFT," *Frontiers in Big Data*, vol. 8, 2025, Art. no. 1677331. DOI: 10.3389/fdata.2025.1677331
3. S. Nwaiwu et al., "The brittleness of transformer feature fusion: A comparative study of model robustness in engineering misinformation detection," *Results in Engineering*, vol. 28, 2025, Art. no. 107783. DOI: 10.1016/j.rineng.2025.107783
4. S. Nwaiwu, N. Jongsawat, and A. Tungkasthan, "Compressed causal reasoning: Quantization and GraphRAG effects on interventional and counterfactual accuracy," 2025. arXiv: 2512.13725 [cs.AI]. URL: https://arxiv.org/abs/2512.13725
5. S. Nwaiwu et al., "Fine-tuned BERT model for hate speech detection in political discourse," 2024. DOI: 10.1109/ICTKE62841.2024.10787162

Skills

Languages	English: fluent (native).
Coding	Python and SQL; Git-based version control.
Tools	PyTorch, scikit-learn, Hugging Face Transformers, Pandas, NumPy, Matplotlib, Jupyter, and Google Colab.
AI / ML	Supervised and unsupervised learning, Random Forest, XGBoost, neural networks, BERT, transformers, LoRA, IA3, ReFT, clustering, and anomaly detection.
Research	Cross-validation, ablation studies, robustness benchmarking, fairness analysis, statistical testing, SHAP, LIME, and reproducible experiments.
Data	Data preprocessing, feature engineering, ETL pipelines, dashboard development, ARIMA, LSTM, and rolling-window evaluation.
Teaching	Python programming, introductory computer science, AI, machine learning, data analytics, visualization, NLP, and digital technology.
Misc.	Academic writing, student mentoring, technical support, and research documentation.
Portfolio	https://github.com/Stevebankz

Miscellaneous Experience

R&D Projects

- **Compressed Causal Reasoning:** Evaluation of quantization and GraphRAG effects on interventional and counterfactual accuracy in language models.
- **Robust and Explainable Hate-Speech Detection:** MSc research combining machine learning, counterfactual reasoning, adversarial evaluation, and fairness analysis.
- **Low-Resource Language Model Adaptation:** Comparative benchmarking of LoRA, IA3, and ReFT for text classification, examining accuracy, parameter efficiency, and training stability.

Miscellaneous Experience (continued)

- **Explainable Fake-News Detection:** Linguistic feature engineering and attribution analysis for transparent classification across multiple domains.
- **Transformer Feature-Fusion Robustness:** Evaluation of misinformation detection models under adversarial perturbations and feature-noise injection.
- **Institutional Analytics Platform:** Undergraduate capstone combining ETL pipelines, institutional data integration, performance monitoring, and interactive visualization.
- **COVID-19 Surge Forecasting:** Time-series pipeline using lag-based features, mobility indicators, and testing data; comparison of ARIMA, Random Forest, and LSTM with rolling forecast evaluation.
- **Gold-Market Volatility Forecasting:** Statistical and machine learning models using macroeconomic and sentiment-derived features, rolling-window evaluation, and feature attribution.
- **Financial Language Model Reliability:** Current doctoral research direction exploring causal evaluation frameworks, interventional benchmarks, and interpretability under changing data distributions.

Awards and Achievements

2022 **TOEIC Excellence Award**, Rajamangala University of Technology Thanyaburi, Thailand.

2021 **ECUBE Excellence Award**, Rajamangala University of Technology Thanyaburi, Thailand.

Certification

- **Google Data Analytics Professional Certificate**, Google.
- **Professional Certification in Data and AI**, IBM.
- **Google IT Support Professional Certificate**, Google.

References

Available on Request